

Lecture FYS5419, April 17, 2024

FYS 5419/9419, APRIL 17, 2024

Phase Estimation algorithm (QPE)

$$U|u\rangle = \lambda|u\rangle$$

$$\lambda = e^{2\pi i \phi} \quad \text{unknown } \phi$$

$$A|u\rangle = \lambda|u\rangle$$

$$\det(A - \lambda I) = 0$$

- 1) Encode the unknown ϕ with a circuit which then

produces a result identical
to a QFT

2) Apply QFT^+ to find ϕ .

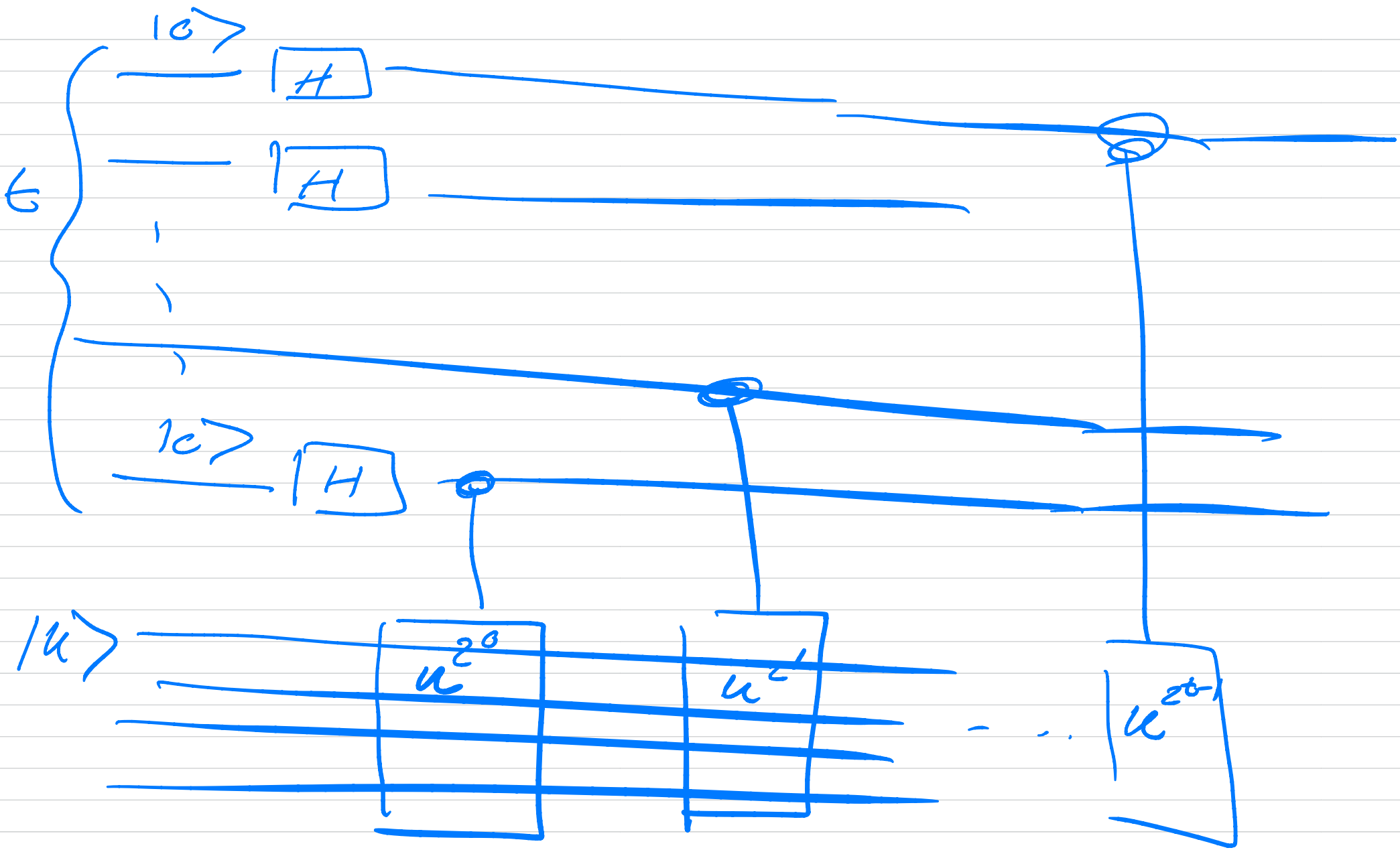
- Define a register with
 t -qubits

$$|101, 12 \dots 1_{n-1}\rangle \rightarrow$$

$$0.1_{n-1} 1_{n-2} \dots 1_0 =$$

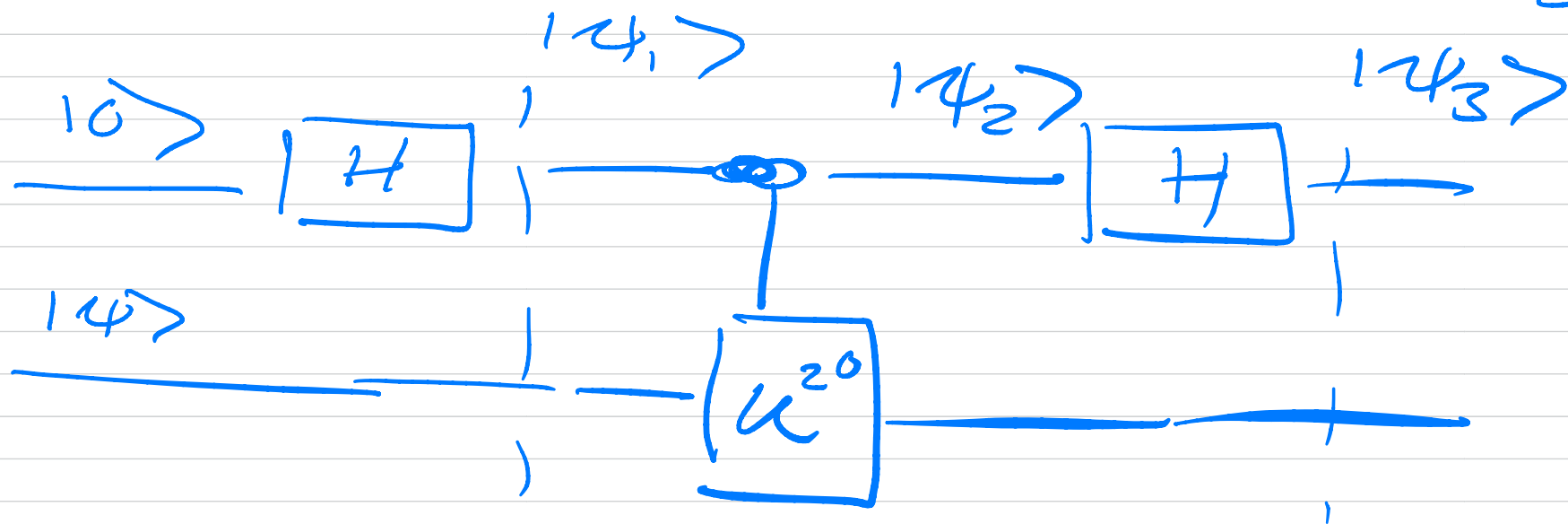
$$\frac{1}{2} 1_{n-1} + \frac{1}{2^2} 1_{n-2} + \dots$$

$$n = t$$



$$\phi = c. \phi_0 \phi_1 \phi_2 \dots \phi_{t-1}$$

$$= \phi_0 \frac{1}{2^1} + \phi_1 \frac{1}{2^2} + \dots + \phi_{t-1} \frac{1}{2^t}$$



$$\lambda = e^{2\pi i 0. \phi_0}$$

$$| \psi_1 \rangle = \frac{1}{\sqrt{2}} (|0\rangle \otimes | \psi \rangle + |1\rangle | \psi \rangle)$$

$$| \psi_2 \rangle = \frac{1}{\sqrt{2}} (|0\rangle | \psi \rangle + e^{2\pi i 0. \phi_0} |1\rangle | \psi \rangle)$$

control
u-gate,

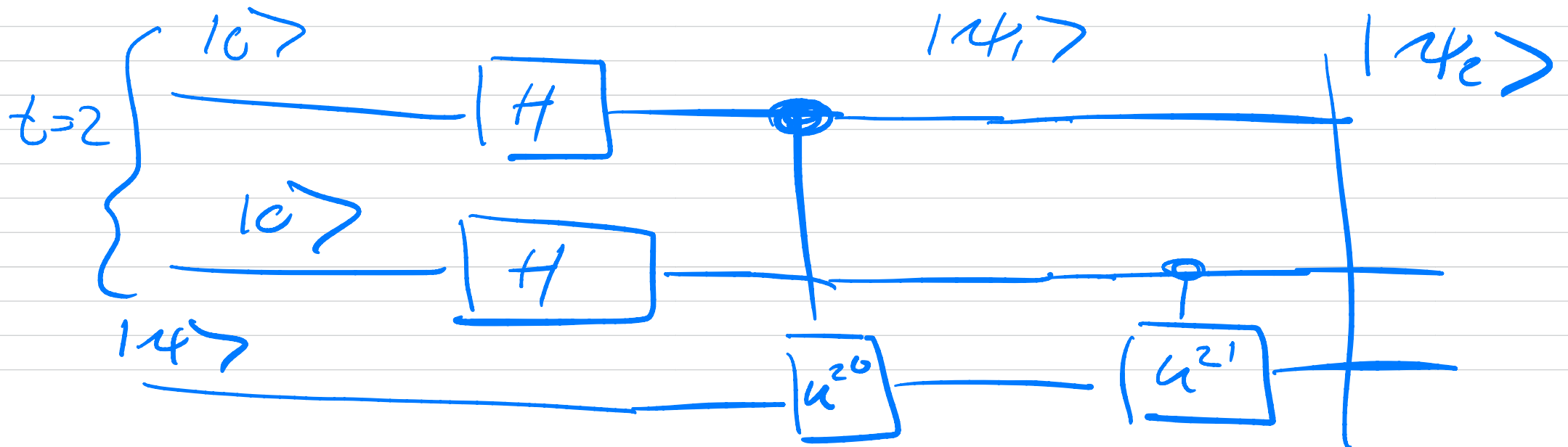
$$= \frac{1}{\sqrt{2}} (|0\rangle + e^{2\pi i 0. \phi_0} |1\rangle) | \psi \rangle$$

$$|\psi_3\rangle = \frac{1}{2} (1 + e^{2\pi i 0. \phi_0}) |0\rangle$$

$$+ \frac{1}{2} (1 - e^{2\pi i 0. \phi_0}) |1\rangle$$

$$\phi_0 = 0 \Rightarrow |\psi_3\rangle = |0\rangle$$

$$\phi_0 = 1 \Rightarrow |\psi_3\rangle = |1\rangle$$



$$|4_1\rangle = \frac{1}{\sqrt{2^3}} \left(\underbrace{|0\rangle + e^{2\pi i 0, \phi_0 \phi_1}}_{1 \text{ qubit}} |1\rangle \right)$$

$$\otimes \left(\underbrace{|0\rangle + |1\rangle}_{2 \text{ qubits}} \right) \otimes |4\rangle$$

$$U^2 |4\rangle = e^{2\pi i 2\phi} |4\rangle$$

$$2\phi = 2(0, \phi_0 \phi_1)$$

$$= 2(\phi_0 z^{-1} + \phi_1 z^{-2})$$

$$= \phi_0 + \phi_1 z^{-1} = \phi_0, \phi_1$$

split fractions at the decim-
mal point

$$e^{2\pi i(2\phi)} = e^{2\pi i(\phi_0 + 0, \phi_1)}$$

$$= e^{2\pi i\phi_0} e^{2\pi i0, \phi_1}$$


$$\phi_0, \phi_1 = \{0, 1\}$$

$$= e^{2\pi i(0, \phi_1)}$$

generalize to

$$e^{2\pi i(z^k \phi)} = e^{2\pi i(0, \phi_k)}$$

For this with $t=2$

$$|\psi_2\rangle = \frac{1}{\sqrt{2^3}} (|0\rangle + e^{2\pi i c_0 \phi_0} |1\rangle)$$

$$\otimes (|0\rangle + e^{2\pi i c_1 \phi_1} |1\rangle) \otimes |4\rangle$$

QFT operation on two
qubits in register (t)

We can apply the two-qubit

QFT[†] operator to retrieve
 $\phi = c_0 \phi_0 \phi_1$

$$FFT^+ | \psi_2 \rangle = | \phi_1 \rangle \oplus | \phi_0 \rangle \oplus | \phi \rangle$$

$$\phi = 0, \phi_0, \phi_1, \phi_2, \dots, \phi_{t-1}$$

$$\frac{1}{2^{t/2}} \left(|0\rangle + e^{i 2^0 \pi \phi} |1\rangle \right) \oplus$$

$$\vdots$$

$$\left(|c\rangle + e^{i \pi 2^{t-2} \phi} |1\rangle \right) \oplus$$

$$\left(|c\rangle + e^{i \pi 2^{t-1} \phi} |1\rangle \right)$$

$$\frac{1}{2^{t/2}} (|0\rangle + e^{i2\pi 0 \cdot \phi_{t-1}} |1\rangle) \otimes$$

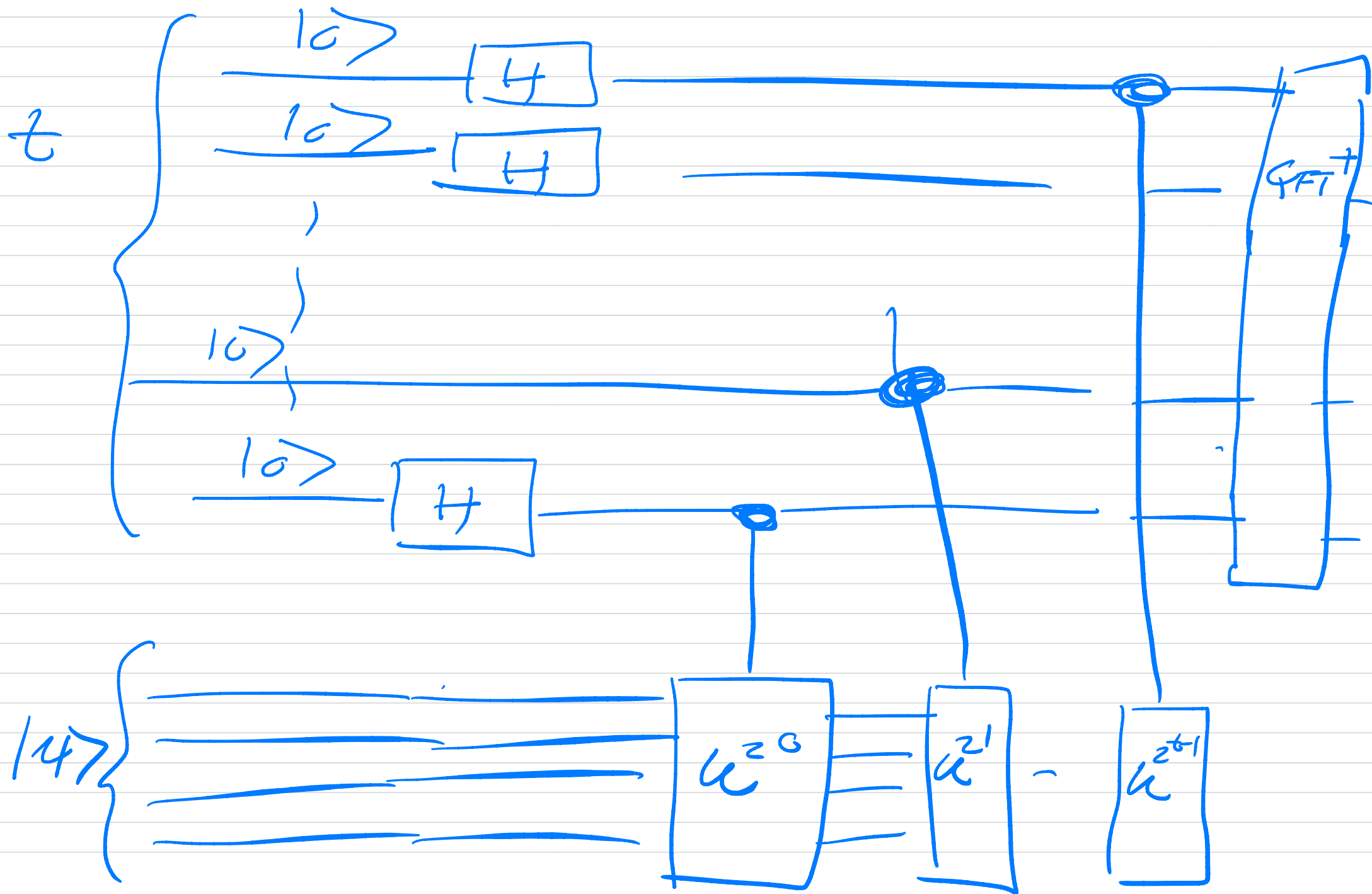
$$(|0\rangle + e^{i2\pi 0 \cdot \phi_{t-1} \phi_{t-2}} |1\rangle) \otimes$$

⋮

$$\otimes (|0\rangle + e^{i2\pi 0 \cdot \phi_{t-1} \phi_{t-2} \dots \phi_1 \phi_0} |1\rangle)$$

$$\phi = 0, \phi_{t-1} \phi_{t-2} \dots \phi_0$$

$$\phi_i = \{0, 1\}$$



Algorithm for QPE

inputs: given number of
qubits for t -register
 U , eigenstate $e^{2\pi i \phi}$

outputs

an t -qubit approximation

to ϕ

Runtime $O(t^2)$ operations

and call to a controlled- U
gate

Procedure:

(i) $|0\rangle \otimes |4\rangle$ initial state

(ii) H on all t -qubits

$$\frac{1}{\sqrt{2^t}} \sum_{j=0}^{2^t-1} |j\rangle \otimes |4\rangle$$

$\{ |j_0 j_1 j_2 \dots j_{t-1}\rangle$

(iii) Control- U on all t -qubits

$$\frac{1}{\sqrt{2^t}} \sum_{j=0}^{2^t-1} e^{2\pi i j \phi} |j\rangle \otimes |4\rangle$$

(iv) apply inverse QFT

(V) measure first register
and extract $\tilde{\phi} \preceq \phi$

Robert Hundt's text,
sec 6.4